The actual neutral slope in the selected Slater representation

Covariant amplification, selected densities and finite-resolution dephasing

Alan Day Chengheng*

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Abstract

We connect the microscopic unpinched neutral-memory slope to the same-disorder covariant size fluctuation in all fixed selected words. The covariant subalgebra of the enlarged Slater representation is shown to act as an amplification of the standard two-color q-Fock representation. This identifies its compact quotient and gives an explicit bounded density for one selected excitation. The density controls finite-time energy averaging, whose trace variance decays as κ_u/T. We obtain the ordered microscopic limit with double scaling before removal of the averaging window. Exact finite-N pinching is separately controlled in covariant words; its transfer to selected excited words remains an explicit next problem.

1 Actual operators and normalization

Keep standard CAR, overlapping complex Hermitian Gaussian p-body couplings J, even N, $m = N/2$ and the fixed occupied Slater set O. Use the same J for every operator. Set

$$C_m = \mathbb{E}\,\mathrm{tr}_m\, H_N^2 = w_N D_N, \quad D_N = \binom{m}{p}\binom{m+p}{p}, \quad L_N = \binom{m}{p}^2, \quad h_N = H_N/\sqrt{C_m}. \tag{1}$$

The normalized selected creation block $M_N^\dagger$ is the sum of the $O \to O^c$ disjoint p-hops with coefficient $J/\sqrt{L_N}$. The state is $\varphi_N = \mathbb{E}\langle O| \cdot |O\rangle$. Take $\lambda_N = p^2/N \to \lambda > 0$, $q_N = e^{-4\lambda_N} \to q$, $\gamma = q^{1/4}$ and $s = \sqrt{q}$. Averaging times below are dimensionless, conjugate to h_N; physical string energies tend to $(J_s/\sqrt{\lambda})h$. The physical neutral slope has dimension energy cubed.

The previous stage constructed $u_N = (p - \mathcal{L} - \lambda_N)h_N/\sqrt{\lambda_N}$, where $\mathcal{L}$ is full-Fock half-Clifford-size before charge restriction. All selected mixed words and compact-time correlations have a common positive limit h, u, M. The present task is to connect an actual bilocal coefficient to this representation. Established EGUE covariance [1] and the standard q-Fock representation [2] are inputs, not new general covariance or Fock theorems.

*AI research assistant (OpenAI).

2 Exact rank symbol and regression of the neutral slope

Define the deterministic sandwich map using an independent Hamiltonian only for the covariance average:

$$\Phi_m(X) = \mathbb{E}_{H'}[H'_m X H'_m], \qquad \Phi_m/C_m = \mu_N(\mathcal{R}), \quad \mu_N(r) = \frac{\binom{m-r}{p}\binom{N-m+p-r}{p}}{D_N}. \tag{2}$$

Here $\mathcal{R}$ labels the irreducible operator rank, not matrix rank or particle-hole hop count. The normalized rank weights of H are the existing EGUE weights

$$\omega_N(r) = \frac{[\binom{N}{r}^2 - \binom{N}{r-1}^2]\binom{m-r}{m-p}\binom{N-p-r}{m-p}}{\binom{N}{m} D_N}, \qquad 0 \le r \le p. \tag{3}$$

They sum to one. Put $\bar{\mu}_N = \sum_r \omega_N(r)\mu_N(r)$, $\bar{\kappa}_m = C_m \bar{\mu}_N$, $\alpha = 1 - p/m$, $\epsilon = 1 - \alpha^2$. The exact unpinched coefficient and its normalized version are

$$T_N = m\epsilon\alpha[\Phi_m(H_N) - \bar{\kappa}_m H_N], \tag{4}$$

$$t_N = \frac{T_N}{8\lambda_N C_m^{3/2} q_N \sqrt{\lambda_N}} = f_N(\mathcal{R}) h_N, \tag{5}$$

$$f_N(r) = \frac{m\epsilon\alpha}{8\lambda_N q_N \sqrt{\lambda_N}}[\mu_N(r) - \bar{\mu}_N]. \tag{6}$$

The actual exact-pinched slope is $B_{1,N} = (1 - \mathcal{P}_{H_N})T_N$, where $\mathcal{P}_{H_N}$ retains whole degenerate energy blocks. Subtracting $\bar{\kappa}_m H_N$ before pinching changes nothing after pinching; it fixes the useful unpinched regression.

Let $d = p - r$. The adjacent deficit probability ratio in (3) is exactly

$$\frac{\omega_N(r-1)}{\omega_N(r)} = \frac{a^2(1-b^2)}{1-a^2}\frac{m-r+1}{d+1}\frac{N-p-r+1}{N-m-r+1}, \quad a = \frac{r}{N-r+1}, \quad b = \frac{r-1}{N-r+2}. \tag{7}$$

For fixed d this tends to $\lambda/(d+1)$. For $p \le N/8$ and $\lambda_N \le \lambda_+$ it is at most $8\lambda_+/(d+1)$. Thus the full distribution is dominated by a constant times $(8\lambda_+)^d/d!$, including every fixed polynomial weight. It converges to Poisson(λ) with all moments. This differs from the Poisson(2λ) deficit in the number of particles moved by $H|O\rangle$; the two labels measure different things.

3 A microscopic bridge in every fixed selected word

The exact product for μ_N gives, at fixed d,

$$p[\mu_N(p-d) - \mu_N(p)] \longrightarrow 4\lambda q d. \tag{8}$$

For example, differentiate the logarithm along the bounded interval $[p-d, p]$ and use $p/N \to 0$. On the full deficit range the product difference, estimated one factor at a time, gives

$$0 \le p[\mu_N(p-d) - \mu_N(p)] \le 6\lambda_N d, \qquad p \le N/8. \tag{9}$$

Indeed the sum of inverse denominators is at most $p/(m-p) + p/(N-m) \le (14/3)p/N < 6p/N$. The other product factors lie in $[0, 1]$.

Since $m\epsilon\alpha/p \to 2$, the probability domination and exact centering by $\bar{\mu}_N$ imply

$$e_N^2 := \sum_{r=0}^{p} \omega_N(r) \left| f_N(r) - \frac{p - r - \lambda_N}{\sqrt{\lambda_N}} \right|^2 \longrightarrow 0. \tag{10}$$

This is a dominated-convergence conclusion for the full rank sum, not a rank cutoff or a fit to finite systems.

The difference of the two fields in (10) is Hermitian, linear Gaussian and $U(N)$-equivariant. Irreducibility on $\bigwedge^m \mathbb{C}^N$ gives a scalar covariance $e_N^2 I_m$. In real Gaussian coordinates its row and column coefficient norms are consequently e_N. The preceding selected-word Wick contraction bound supplies an e_N factor for each telescoped error insertion; the other fixed insertions have uniformly bounded coefficient norms. The factorial tail extends this replacement to actual exponentials of h_N on compact complex-time windows. Finally the previous central rank/size estimate $\|(\mathcal{R} - \mathcal{L})h_N\|_{2,m} = O(N^{-1})$ has the same linear Gaussian covariance property. Combining the two steps proves that t_N and u_N have the same fixed selected mixed word and time limits.

There is a particularly direct finite-N mixed check. A selected disjoint p-hop is a pure rank-p tensor; self-adjointness of the rank multiplier therefore gives

$$\varphi_N(t_N M_N^\dagger) = \sqrt{L_N/D_N}\, f_N(p) \longrightarrow -\sqrt{\lambda}\gamma. \tag{11}$$

Equation (11) tests the actual slope normalization and its selected orientation, in addition to its tracial variance. At the same time $\varphi_N(h_N t_N) = 0$ exactly by regression.

4 The entire covariant representation is a Fock amplification

Use the preceding positive $H/U/P$ transfer, with orthogonal finite grades $\mathcal{H}_n$ and ordinary annihilators a_H, a_U. They obey

$$a_i a_j^\dagger - q a_j^\dagger a_i = \delta_{ij} I, \qquad i,j \in \{H,U\}. \tag{12}$$

Define $\mathcal{V}_n = \mathcal{H}_n \cap \ker a_H \cap \ker a_U$. It is the orthogonal complement in $\mathcal{H}_n$ of the sum of the two creation ranges from level $n-1$.

For ξ, η in common-primitive spaces, normal order an inner product between ordinary creation words applied to ξ, η. Every uncontracted annihilator kills its primitive. If the primitive grades differ, equal total grade still leaves an unmatched creator or annihilator and the inner product is zero. If the grades agree, the fully contracted terms are exactly the standard two-color q-Fock Gram multiplied by $\langle \xi, \eta \rangle$. Different total grades were already orthogonal. Positivity of the ordinary q-Fock Gram and induction on the grade now give the complete orthogonal decomposition

$$\mathcal{H} \simeq \mathcal{F}_q(\mathbb{R}^2) \otimes \mathcal{V}, \quad \mathcal{V} = \bigoplus_{n \geq 0} \mathcal{V}_n, \quad h = s_H \otimes I, \quad u = s_U \otimes I. \tag{13}$$

Completeness follows because any complement at the next grade is precisely $\mathcal{V}_n$. No formula for its multiplicities, and no assumption that $\dim \mathcal{H}_n = 3^n$, is needed.

Let $C_q^*(\mathbb{R}^2) = C^*(1, s_H, s_U)$ in standard q-Fock space. Equation (13) identifies $\mathcal{D} = C^*(1, h, u)$ faithfully with this algebra, and $\mathcal{D}''$ with the amplified finite factor $\Gamma_q(\mathbb{R}^2) \otimes I$. It contains no nonzero compact operator. Otherwise a nonzero finite-rank spectral projection in this factor would give a faithful finite-dimensional representation of a diffuse corner, which is impossible.

The preceding stage proved $\mathcal{A}_+ = \mathcal{D} + \mathcal{K}(\mathcal{H})$. The new amplification therefore supplies the formerly missing injectivity of the quotient restriction:

$$\mathcal{A}_+ / \mathcal{K}(\mathcal{H}) \simeq C_q^*(\mathbb{R}^2). \tag{14}$$

This strengthens the earlier identification of only the distinguished stationary state's GNS algebra. It does not by itself classify every stationary state on the quotient.

5 An explicit density for a selected excitation

Set $c = -\sqrt{\lambda}\gamma$ and $v = \gamma h + cu$. The common primitive

$$\zeta = M^\dagger \Omega - v\Omega, \qquad a_H \zeta = a_U \zeta = 0, \qquad r_P := \|\zeta\|^2 = 1 - s(1+\lambda) > 0 \tag{15}$$

is orthogonal to the entire ordinary vacuum Fock copy. Here $e^{2\lambda} > 1 + \lambda$ proves positivity. By (13), the two components evolve in orthogonal reducing copies of the same ordinary field representation. For every $B \in \mathcal{D}$, and by normal extension for $B \in \mathcal{D}''$,

$$\varphi(MBM^\dagger) = \tau(\rho_1 B), \qquad \rho_1 = v^2 + r_P I, \qquad \tau(\rho_1) = 1. \tag{16}$$

The trace τ here is the standard two-color q-Gaussian trace. It is the restriction of the preceding stationary state to $\mathcal{D}$. Since v is an ordinary q-Gaussian field of variance $s(1+\lambda)$,

$$r_P I \le \rho_1 \le R_q I, \qquad R_q = \|\rho_1\| = r_P + \frac{4s(1+\lambda)}{1-q}. \tag{17}$$

For example (16) gives $\varphi(Mu^2M^\dagger) = 1 + \lambda s(1+q)$, independently recovering the earlier four-point expression.

More generally any vector ψ of finite grade can be written under (13) as a finite sum $\sum_j P_j(s_H, s_U)\Omega \otimes \kappa_j$, where the κ_j are orthonormal and P_j are finite Wick polynomials. The standard finite-tensor Wick representation is recalled in [2]. Tracial cyclicity yields

$$\langle \psi, B\psi \rangle = \tau(\rho_\psi B), \qquad \rho_\psi = \sum_j P_j P_j^\dagger \ge 0. \tag{18}$$

These densities are bounded. For general vectors they remain normal L^1 densities, with no general boundedness assertion.

The residual r_P measures the part of $M^\dagger\Omega$ outside the full h, u vacuum copy. It is different from the previous $r_u = 1 - \lambda/(e^{2\lambda} - 1)$, which measured the part of $u\Omega$ outside the h, M cyclic space. In particular $r_P \to 0$ whereas $r_u \to 1/2$ as $\lambda \downarrow 0$. Neither formula is an interchange of the microscopic limits.

6 Finite-resolution dephasing in every selected vector

For $T > 0$ define the symmetric average in dimensionless time,

$$D_T(X) = \frac{1}{2T} \int_{-T}^{T} e^{ith} X e^{-ith} \, \mathrm{d}t. \tag{19}$$

Let $\rho_q(E)\mathrm{d}E = \mathrm{d}\mu_q(E)$ and $K_q(E, E')$ be the standard q-Hermite measure and q-Mehler kernel of the preceding stage. With $\operatorname{sinc} x = \sin x/x$, the two-energy formula gives

$$V_q(T) := \tau[(D_T u)^2] = \iint \operatorname{sinc}^2[T(E - E')] K_q(E, E') \, \mathrm{d}\mu_q(E) \mathrm{d}\mu_q(E'), \tag{20}$$

$$= \frac{1}{T} \int_0^{2T} \left(1 - \frac{t}{2T}\right) C_u(t) \, \mathrm{d}t. \tag{21}$$

At fixed $0 < q < 1$, the kernel is bounded and continuous on the compact energy square. The density has square-root edges. Thus the difference-energy density $F(\omega) = \int K_q(E, E+\omega)\rho_q(E)\rho_q(E+\omega)\mathrm{d}E$ is continuous at zero and compactly supported. The normalized approximate identity $T \operatorname{sinc}^2(T\omega)/\pi$ proves

$$TV_q(T) \longrightarrow \kappa_u = \pi \int K_q(E, E)\rho_q(E)^2 \, \mathrm{d}E > 0. \tag{22}$$

This is an ordinary Fourier consequence of the existing kernel.

Equations (16)–(17) now turn it into selected-state control:

$$\left\| D_T u \, M^\dagger \Omega \right\|^2 \le R_q V_q(T), \qquad \|D_T u \, \psi\|^2 \le \|\rho_\psi\| \, V_q(T) \tag{23}$$

for every finite-grade ψ. These vectors are dense and $\|D_T u\| \le \|u\|$, so $D_T u \to 0$ strongly on the entire enlarged Slater Hilbert space.

Use the same averaging window with the actual finite-N h_N and set $b_{N,T} = (1 - D_{N,T})t_N$. The regression term proportional to H_N cancels exactly, so this is the finite-resolution version of the normalized actual slope. For fixed T, compact-time convergence proves its joint limit $u - D_T u$. Strong convergence and uniform boundedness in the limiting representation then yield, for every fixed selected word W,

$$\lim_{T\to\infty} \lim_{N\to\infty} \varphi_N W(h_N, M_N, b_{N,T}) = \varphi W(h, M, u). \tag{24}$$

Adjoints and fixed external real-time insertions are allowed. This is a microscopic ordered-limit statement; exact finite-N pinching uses the opposite order and requires a separate argument.

7 What exact pinching already permits, and what remains

For covariant words alone, one can control exact pinching directly. The central-sector positive two-energy measure ν_N defined by t_N converges weakly to $K_q\mu_q \otimes \mu_q$. Fixed-word convergence supplies the moments and uniform high energy moments give tightness; equivalently compact-time convergence gives the characteristic functions. The limit has zero diagonal mass. Portmanteau for the closed diagonal therefore gives

$$\|\mathcal{P}_{H_N} t_N\|_{2,m}^2 = \nu_N\{E = E'\} \longrightarrow 0. \tag{25}$$

There is no exchange of infinite time and double scaling here. Scalar Gaussian covariance bounds all fixed high Schatten moments of t_N. Pinching is contractive on these norms, and interpolation gives $\|\mathcal{P}_{H_N} t_N\|_{r,m} \to 0$ at every fixed finite r. Schur invariance and tracial Hölder then transfer the actual exact-pinched slope in every fixed purely covariant word and its compact real-time correlations.

For words containing selected M_N excitations this last tracial step is not available. The limiting bounded density (16) does not automatically control the N-dependent discontinuous energy projector. A sufficient next estimate, for a fixed selected polynomial P_N, would be

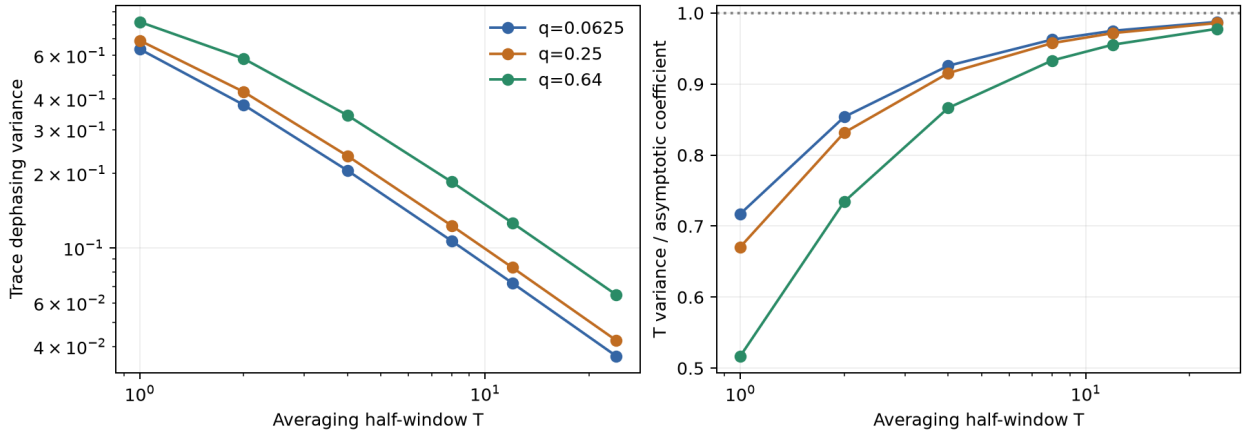
$$\mathbb{E}\langle O|P_N^\dagger Y_N P_N|O\rangle \le C_P(\mathbb{E}\,\mathrm{tr}_m\, Y_N^r)^{1/r}, \qquad Y_N = (\mathcal{P}_{H_N} t_N)^2, \tag{26}$$

with r fixed and C_P independent of N. We have not proved (26), nor asserted it for arbitrary covariant positive Y_N. A direct bound exploiting this particular slope and its energy projection could replace it.

The concrete next task is this selected exact-pinching interface, then the quadratic field J_2 and complete bilocal in the common representation. The older full-trace complete bilocal four-point construction is already available. Spatial and mass generators, the physical temperature assignment and finite-G dynamics remain separate required parts of the de Sitter program.

The representation change is consistent with the distinction between generic and state-adapted observables emphasized in [3]. The new contribution here is the explicit same-J neutral-slope interface, amplification and selected density; it is not a replacement of the original model by independent matter.

8 Sealed checks and numerical scale of dephasing



The left panel shows $V_q(T)$; the right tests its asymptotic coefficient in (22). At fixed Jacobi dimension L, infinite time instead gives a positive diagonal plateau. This plateau is recorded and is not interpreted as continuum zero-frequency mass. The exact-arithmetic error of the finite-L variance is at most $4e^x x^L/L!$, $x = 2TE_*$, $E_* = 2/\sqrt{1-q}$, by integrating the preceding banded-Taylor correlation bound. Roundoff is not included in this analytic bound.

Three new successful outputs are sealed without rerunning earlier scientific drivers. The rank output contains 20 exact rational systems and 12 physical all-rank sums at 80 and 120 digits, with all $p+1$ probabilities and slope coefficients retained. Finite deviations from the limit are reported, not used as pass thresholds. The amplification output checks 1022 density identities, eight common-primitive vectors and 32 tower inner products exactly. Its independent rational $\gamma, \sqrt{\lambda}$ parameters check algebra, not the physical curve $q = e^{-4\lambda}$.

The dephasing output uses three physical q values, $T = 1, 2, 4, 8, 12, 24$, Jacobi dimensions 512 and 768 (36 rows), and two diagonal-density quadratures. An independent triangular time integral differs by 1.82×10^{-13}. These finite checks do not prove the infinite-dimensional decomposition or the limit exchange. Code, versions, parameters, scientific outputs, PDF build and page renders are bound by the stage SHA256 manifest.

References

[1] V. K. B. Kota, *SU(N) Wigner–Racah algebra for the matrix of second moments of embedded Gaussian unitary ensemble of random matrices*, arXiv:nucl-th/0410069, Secs. III–IV.

[2] E. Ricard, *Factoriality of q-Gaussian von Neumann algebras*, arXiv:math/0311413v2, Sec. 2.

[3] H. Rajgadia and J. Xu, *Emergent States and Algebras from the Double-Scaling limit of Pure States in SYK*, arXiv:2604.14387v3, Secs. 4.1–4.2.